

Resetting the Clocks: Mitochondrial Quality Control via Genetic Adaptation

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Summer Research Academies

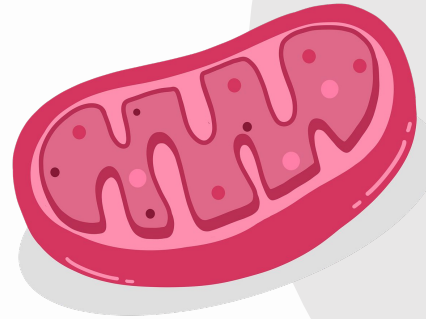
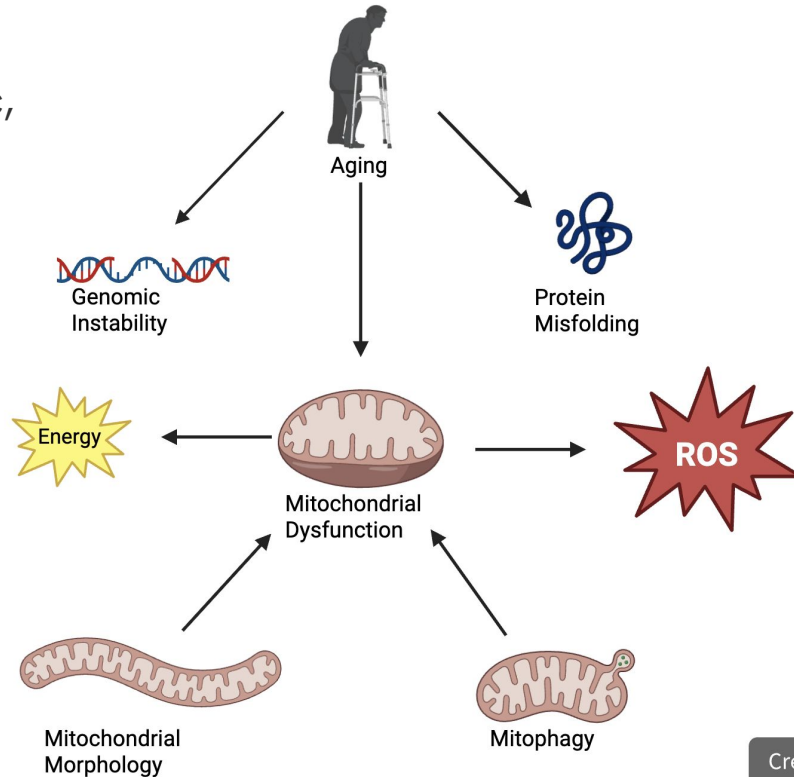


Image: Pixabay
(<https://pixabay.com/illustrations/mitochondrion-cell-biology-6258212/>)

Mitochondrial Disease & Aging

- According to Cleveland Clinic, 1 in 5000 are affected by genetic mitochondrial diseases.
- In just 13 years, the prevalence of adult mitochondrial diseases in Southwest Finland doubled from 4.6 to 9.2 per 100,000.



Fission

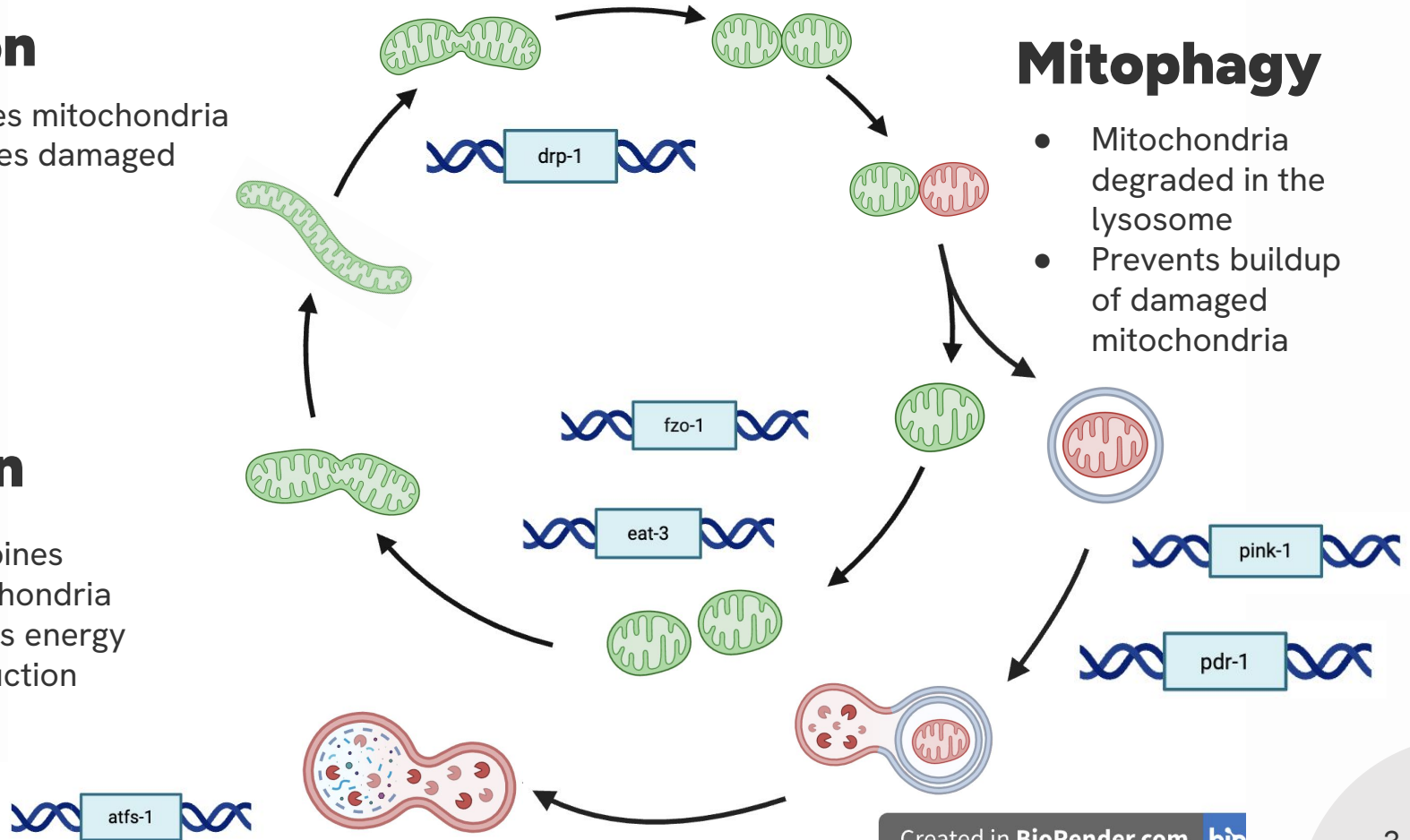
- Divides mitochondria
- Isolates damaged parts

Fusion

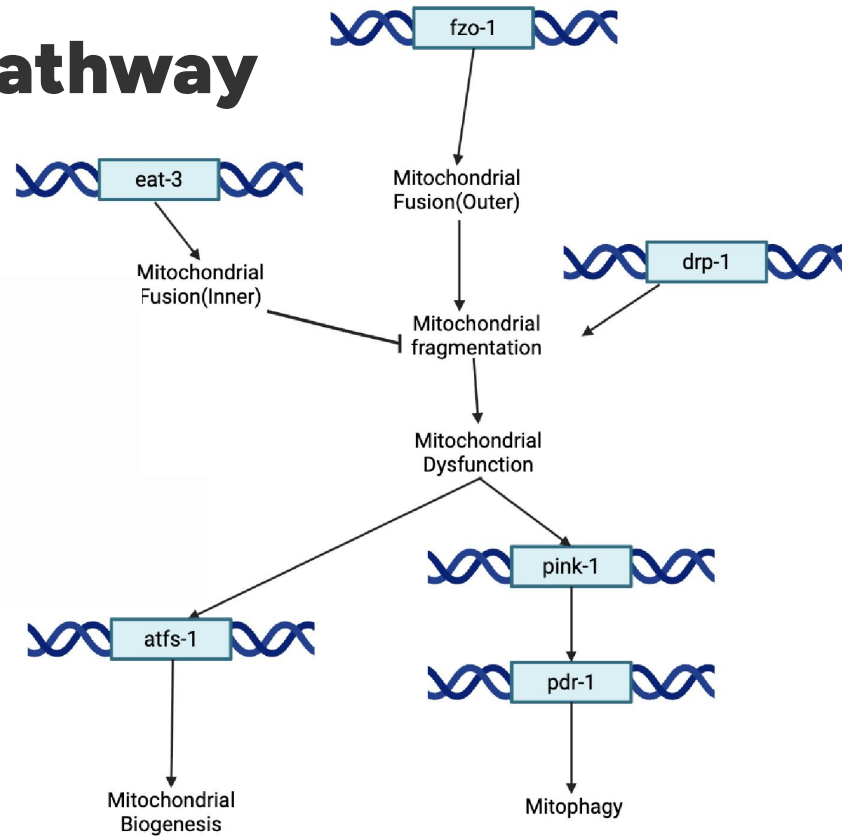
- Combines mitochondria
- Boosts energy production

Mitophagy

- Mitochondria degraded in the lysosome
- Prevents buildup of damaged mitochondria



Genetic Pathway



Model Organism: *Caenorhabditis elegans*

- Why *C. Elegans*?

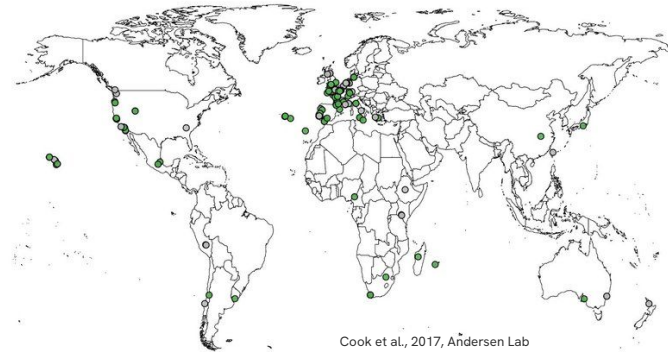
- Simple anatomy
- Short life span
- Hermaphrodite
- Well studied genome
- Human orthologs



Hardin Lab, University of Wisconsin-Madison

- Previous studies have looked at the *N2* *C. elegans* strain but have not studied wild type strains *CX11314* and *EG4725*.

- Scientists have also identified the genes that play a role in maintaining mitochondrial health.



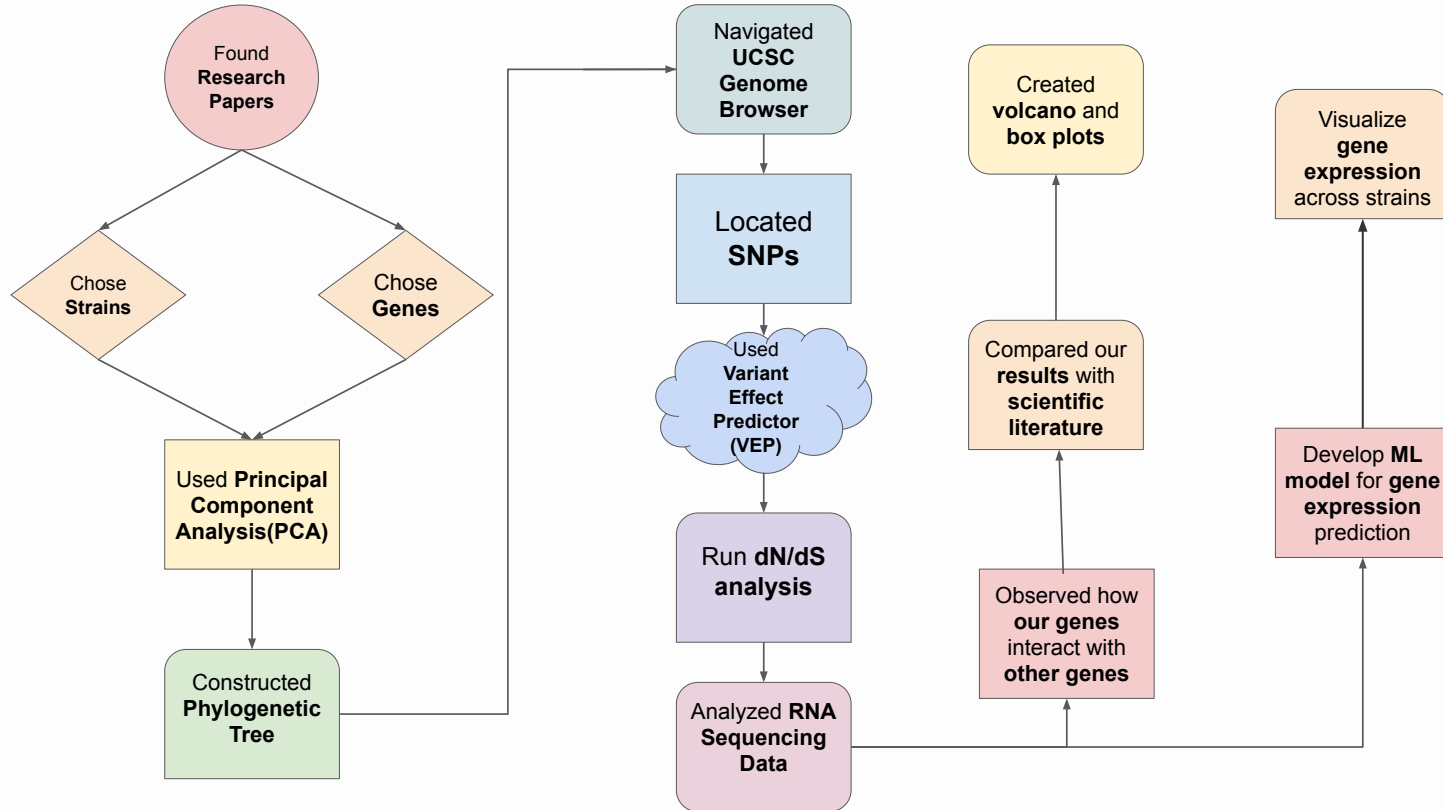
Cook et al., 2017, Andersen Lab

Question

Research Question: What **genetic variants** in **mitochondrial morphology** and **mitophagy-related genes** contribute to the regulation of **mitochondrial quality and heteroplasmy** in the *C. elegans* strains **CX11314** and **EG4725** compared to the **N2** reference strain?

Hypothesis: We hypothesize that **natural genetic variation** in **mitochondrial morphology** and **mitophagy-regulating genes** enhances **mitochondrial quality** control in the wild *C. elegans* strains **CX11314** and **EG4725**, leading to reduced mitochondrial dysfunction and lower levels of **heteroplasmy** compared to the **N2** reference strain.

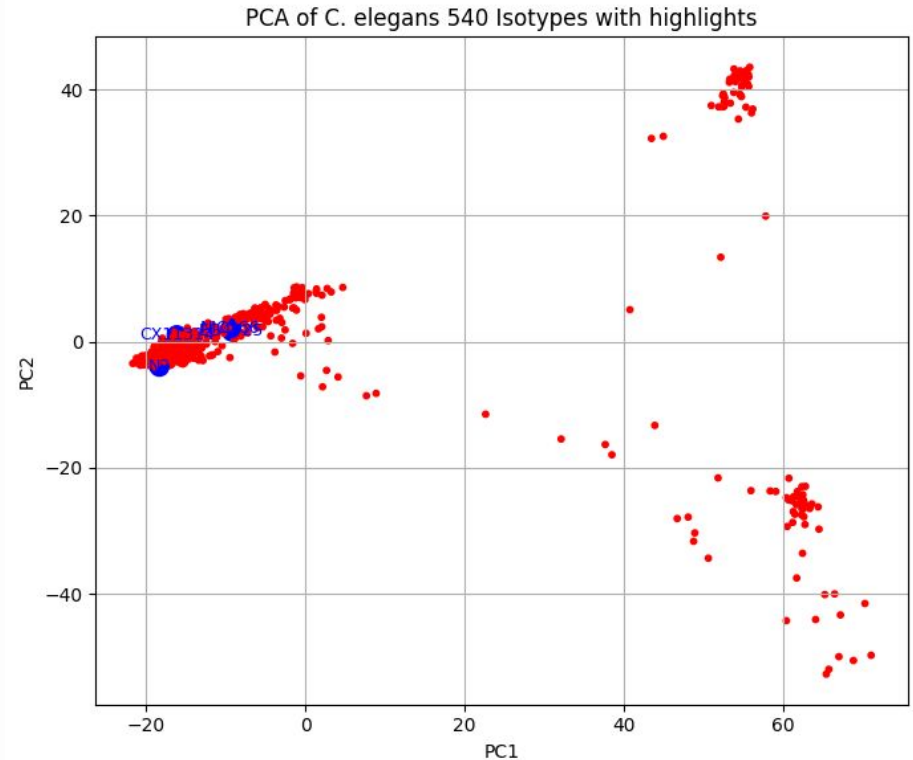
Methodology



Principal Component Analysis (PCA)

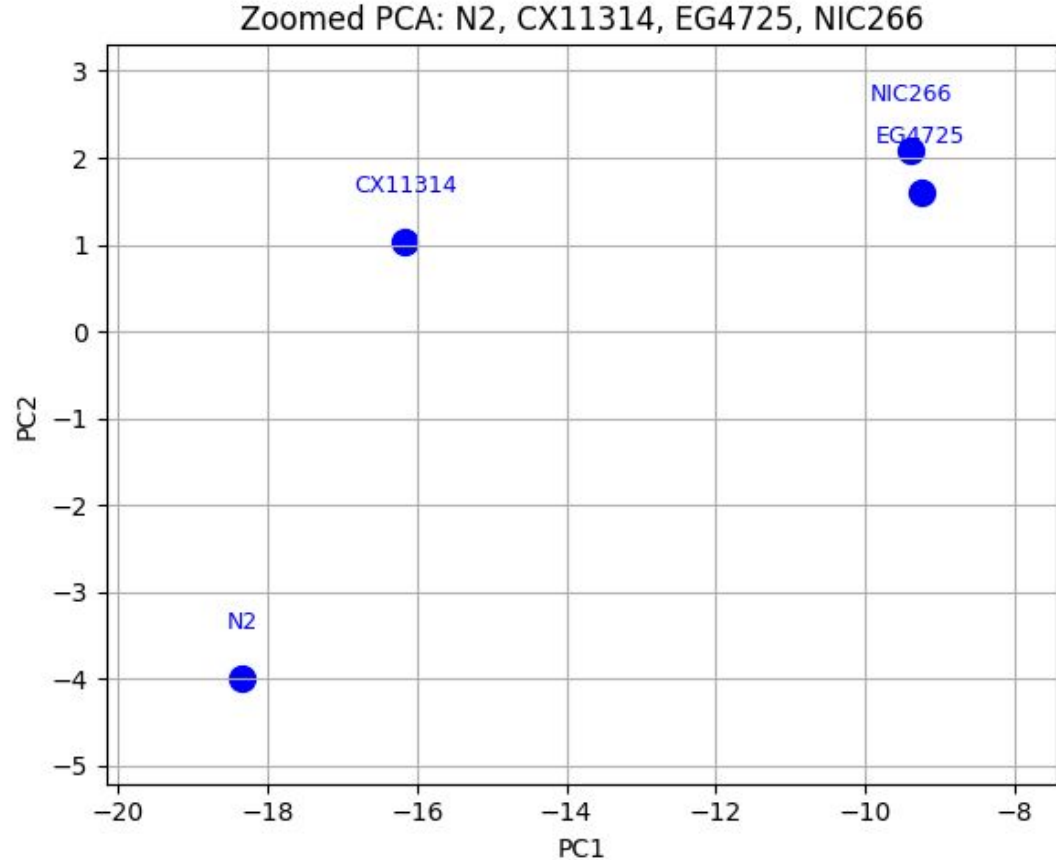
used to simplify complex, high-dimensional data.

- Easier visualization and interpretation
- Illustrates the genetic clustering of strains
 - Distance between points is proportional to the genetic distance between strains.



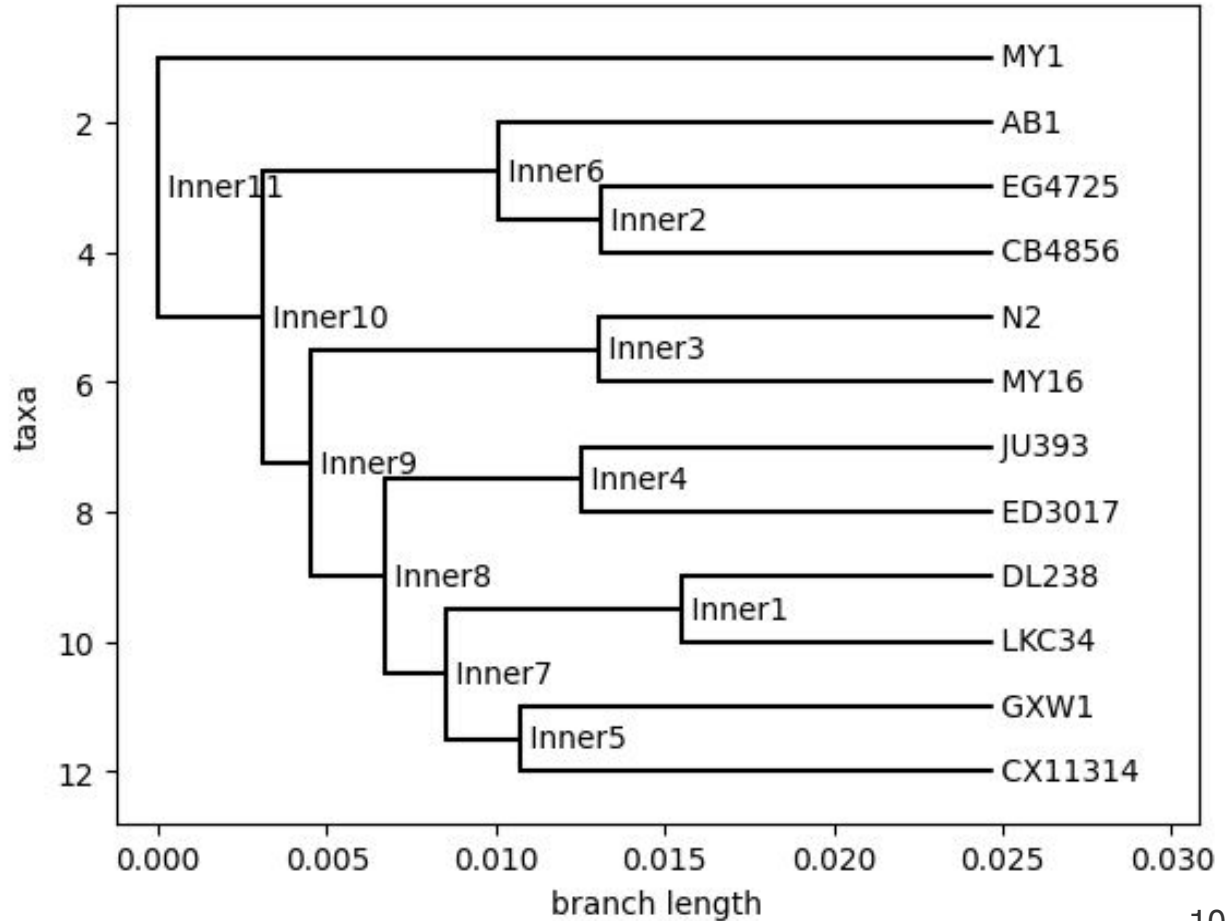
PCA Analysis

- Shows how *CX11314* and *EG4725* are genetically divergent from *N2*.
- *NIC266* is the genetically closest to *EG4725*
 - Biological sense to use *NIC266* as a proxy for *EG4725*.



Phylogenetic Tree

- Like a family tree
- Branch lengths = distance
- Our strains far from N2



Ratio of non-synonymous to synonymous substitutions (dN/dS):

- Genes can:
- Be preserved
 - Change randomly
 - Adapt to environmental changes

Gene	dN	dS	dN/dS
atfs-1	34	41	0.829
pink-1	5	5	1
drp-1	12	20	0.6
pdr-1	15	19	0.789
eat-3	5	5	1
fzo-1	5	5	1

Variant Effect Prediction (VEP)

- Near 1:1 ratio
 - Moderate evolutionary pressure.
- Highest Synonymous variants: *drp-1*
 - Crucial in mitochondrial fission
- Highest Missense variants: *eat-3*, *atfs-1*
 - Adaptive evolution under stress
- Stop gained Variation in *pdr-1*
 - Truncated proteins and altered functions.

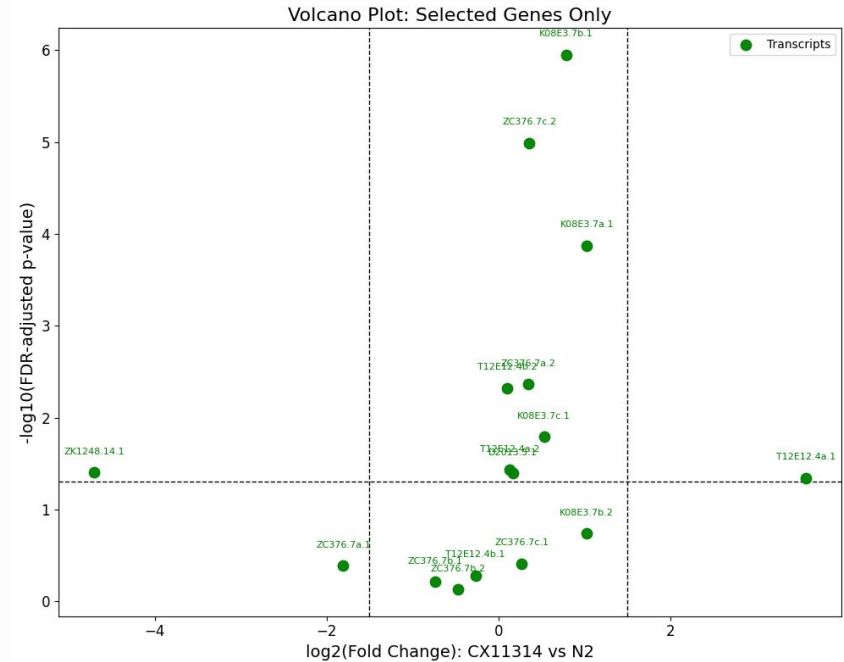
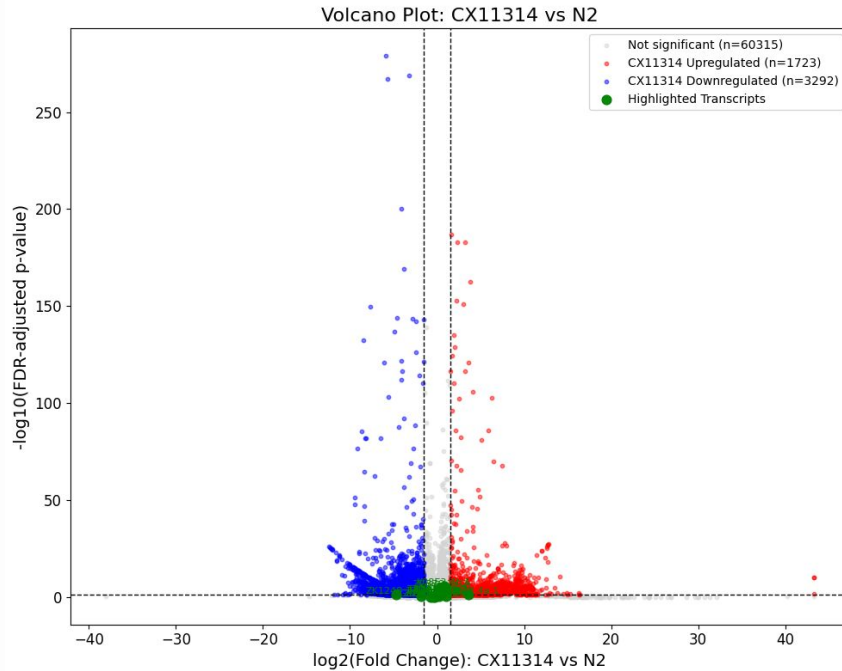
Genes	Synonymous Variants	Missense Variants	Stop Gained
fzo-1	56%	44%	0
eat-3	52%	48%	0
drp-1	75%	25%	0
pink-1	59%	41%	0
atfs-1	55%	45%	0
pdr-1	53%	43%	3%

Protein Differences

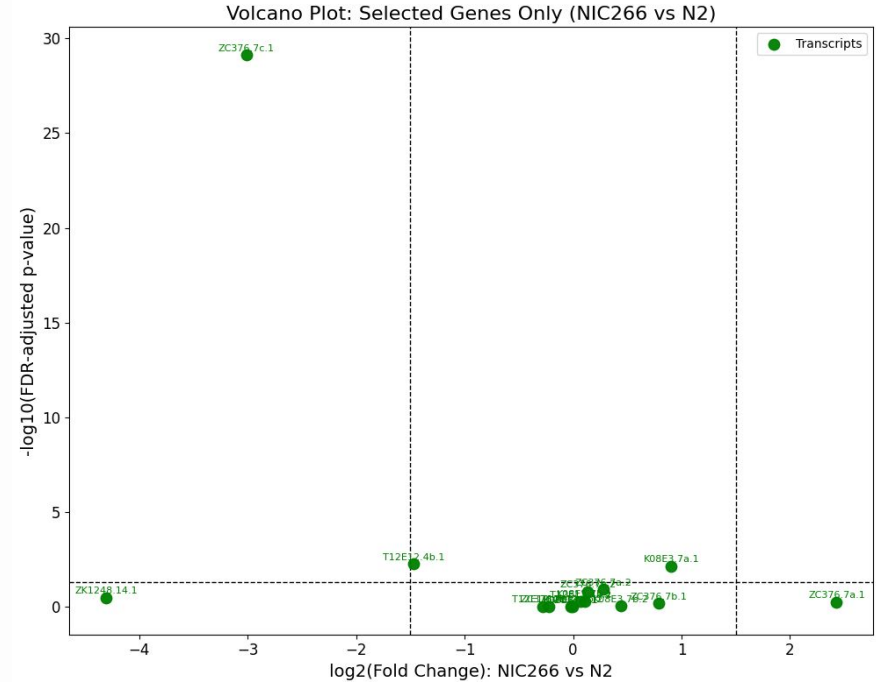
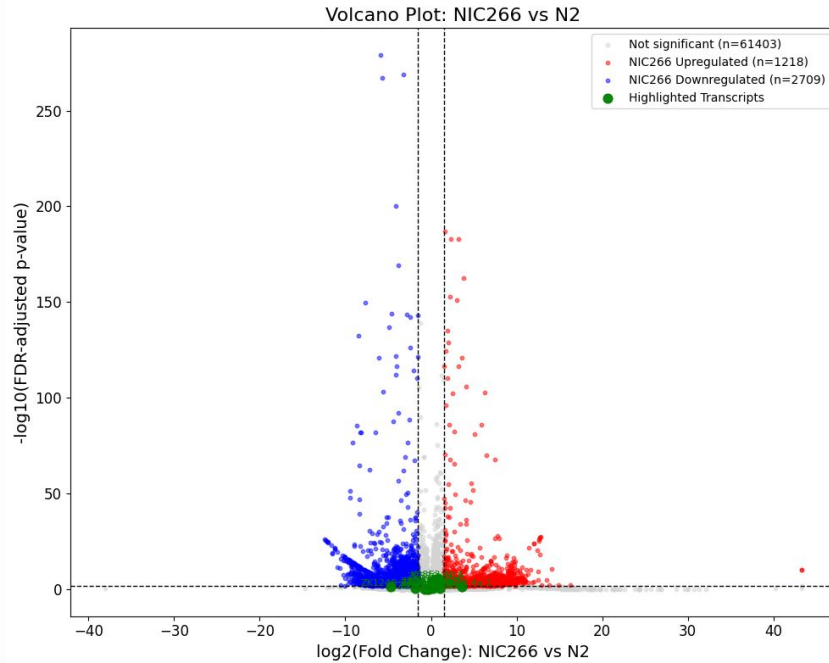
- *fzo-1* and *eat-3* had moderate variations.
- *drp-1* had conservative changes
- *pdr-1* had diverse changes
- *atfs-1* had the most variants
 - Could suggest rapid evolution

Gene	Number of Variants	Specific variants
fzo-1	5	1 P->S, 1 R->Q, 1 A->T, 1 V->M, 1 L->I
eat-3	5	1 I->M, 1 A->P, 1 K->R, 1 V->A, 1 A->T
drp-1	12	4 K->R, 4 L->I, 4 T->I
pdr-1	15	3 S->T, 4 E->K, 4 G->D, 2 L->R, 2 F->L
atfs-1	34	7 P->L, 7 Q->R, 7 S->T, 6 V->A, 7 Y->N
pink-1	5	1 E->K, 1 A->S, 1 P->S, 1 V->M, 1 N->D

Volcano Plot

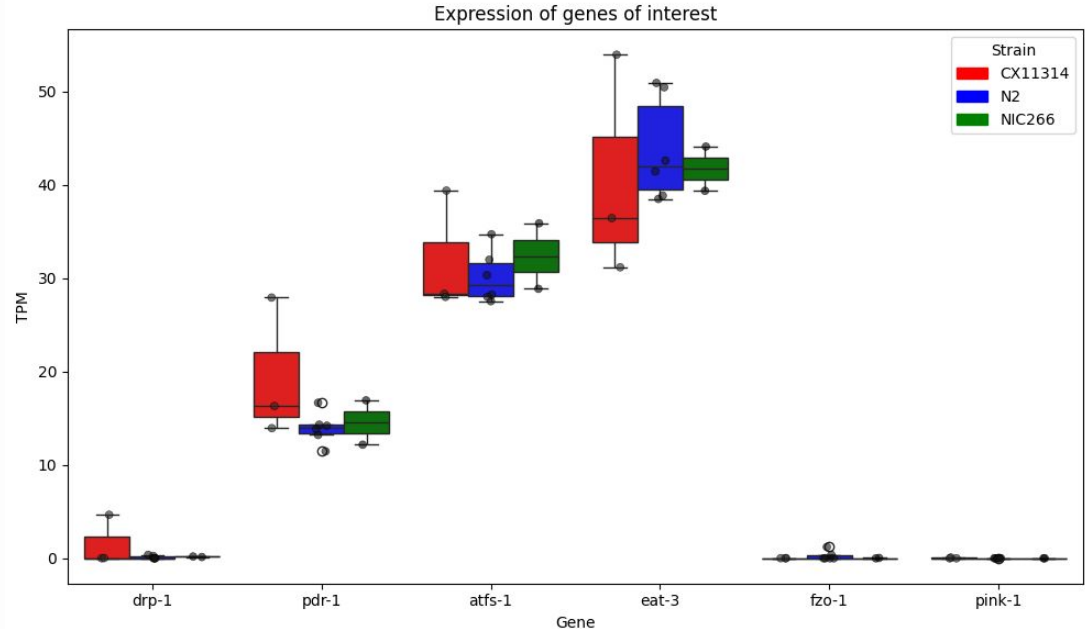


Volcano Plot



Expression of Genes

- *eat-3* had the highest expression overall.
- *pdr-1* had the highest variability in *CX11314*.
- However, there was no statistically significance in expression levels.
 - Suggests that these genes are for fine-tuning mechanisms.



Prediction of *EG4725* expression via ML

Data Preparation:

1. Filtered VCF and TPM dataset to remove null values.
2. Log2-transformed and z-score normalized expression data.
3. Filtered to only include genes of interest

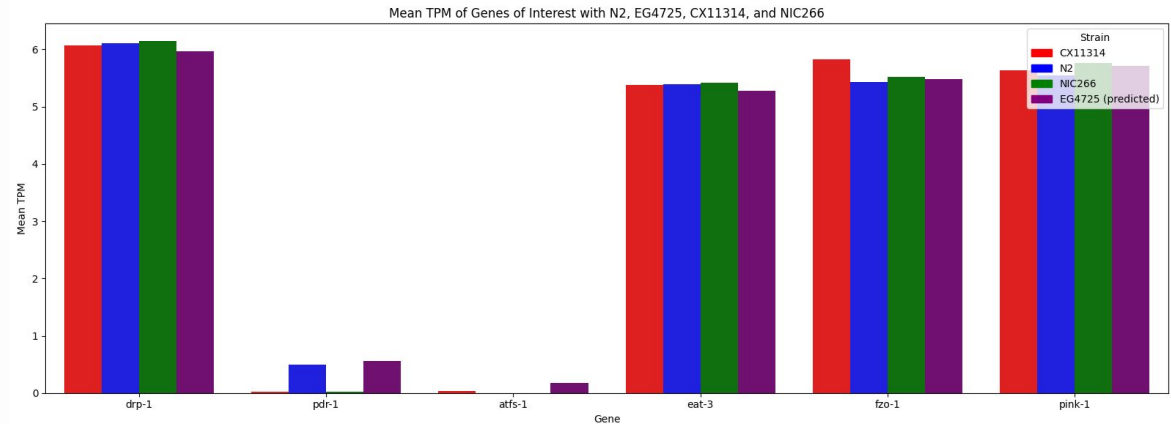


Implementing Model

1. Trained the ML model on 204 strains
2. **Neural Network Classifier:**
 - Input: expression/SNP data
 - Output: predicted strain label
 - Used Leave One Out Cross Validation(LOOCV) to test generalization.
3. **Spearman Correlation:**
 - To test how well gene expression tracks with strain identity
 - $R^2 = 0.959$; very strong correlation.

Visualization of Predicted Expression

- The predicted expression for *EG4725* follows the trend with the expression of *N2*, *CX11314*, and *EG4725*.
- Interestingly, *EG4725* had higher expression for *pdr-1* and *atfs-1*.



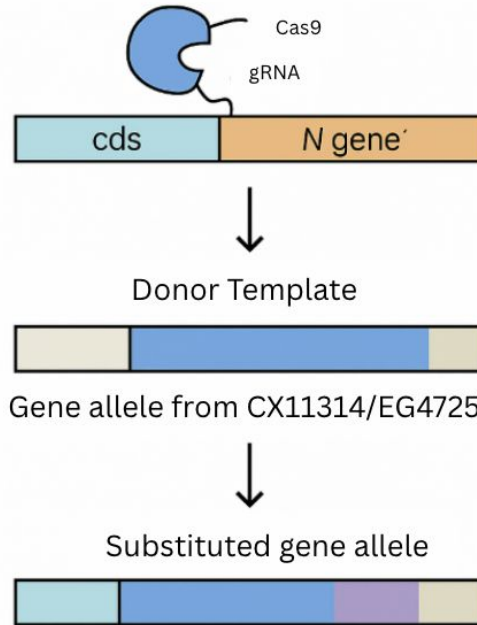
Conclusion

- Our hypothesis was correct
- Natural variation in our strains enhanced mitochondrial function
- *CX11314* and *EG4725* adapted over time for enhanced mitochondrial control
- PCA plot and phylogenetic tree support this
- RNA Seq also supports this
- Small changes over time lead to noticeable differences

Proposal

- Replace *N2* gene allele with those from *CX11314* and *EG4725*
- Swap promoters and coding sequences of *pdr-1* and *atfs-1* from *CX11314/EG4725* into *N2*.
- Experimental comparisons:
 - *N2* (control)
 - *N2* with *pdr-1* EG allele
 - *N2* with *atfs-1* EG allele
 - *EG4725* (baseline)

CRISPR-Cas9 mediated Homology Directed Repair (HDR)



Precisely substitute *N2* gene alleles with those from *CX11314/EG4725*

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Questions?

